

Gen-AI training software installation and lab setup guide

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Overview

This document describes the setup needed on the participant PC for Gen AI training

1. URL Whitelisting

Following URLs need to be accessible from participant's laptop:

- 1) MS Azure portal:
<https://portal.azure.com>
- 2) Google Colab:
<https://colab.research.google.com>
- 3) Kaggle:
<https://kaggle.com>
- 4) Huggingface:
<https://huggingface.co/>

- 5) LM Studio
<https://lmstudio.ai/>
- 6) Pypi (for python packages installation):
<https://pypi.org/>
- 7) Access to at least one AI chatbot such as:
OpenAI ChatGPT: <https://chatgpt.com>
Microsoft copilot: <https://copilot.microsoft.com>
Google Gemini: <https://gemini.google.com>
Anthropic Claude (Amazon backed): <https://claude.ai> (good for code generation)
- 8) Github:
<https://github.com>
- 9) Microsoft Onedrive or Google Drive (optional)
- 10) Zapier:
<https://zapier.com/>
- 11) N8n
<https://n8n.io>
- 12) Comet:
<https://www.comet.com/site/>

2. Python

1. Install Python

<https://www.python.org/downloads/release/python-3127/>

Use Python 3.12.6 or 3.12.7 only as it is tested with the libraries we will work on
DO NOT INSTALL LATEST PYTHON VERSION

When installing make sure the “Add python to system path” checkbox is selected

2. Test python

Open a command prompt and give the following command:

```
python -version
```

It should show the python version, namely, 3.12.7

If it errors finding the executable, add python.exe location to System environment variable: PATH

You can find the location of python.exe by opening the windows menu, right click on Python app (open file location), copy path

3. VS Code

1. Download and install [VS Code](https://code.visualstudio.com/)

<https://code.visualstudio.com/>

2. Install python extensions in VS Code

1. Open VS Code
2. Install the Python extension:
 - Click the Extensions icon in the sidebar (or press Ctrl+Shift+X)
 - Search for "Python"
 - Install the Microsoft Python extension
 - Other VS Code extensions that are needed only if you want to run the Python notebooks in VS Code (you can always run them in colab)
Jupyter, Jupyter Notebook renderers, ipykernel

Create a Virtual Environment

Understanding Project Root folder

Project root folder (such as **genaitraining**) or any folder in which you are putting your code, provides the base folder to work against.

When opening VS Code, this is the folder you will open (not any other subfolder)

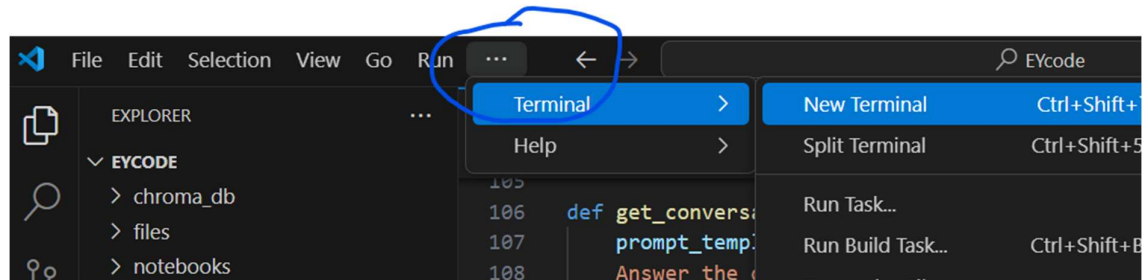
This is important as your project virtual environment, private configuration (.env) will all reside in your project root folder.

Hence, whenever starting VS Code, **always open the project root folder and not any other folder**

Create a virtual environment for this training

1. Create a new folder for the labs for this training, called **genaitraining (or code)** in your home directory or in Desktop
2. Open the folder in VS Code (File > Open Folder)

3. Start a terminal



4. In the terminal window, give the following commands to create a new virtual environment:

```
python -m venv venv
```

```
venv\Scripts\activate
```

You will see a popup asking you to use this virtual environment for your project.
Click on “Yes”

Install packages needed for Gen AI Labs

Python package manager **pip** allows you to install all needed packages from a file commonly named as **requirements.txt**

We will create requirements.txt and put all the packages we need for Gen AI development in it. We will then install those packages in our virtual environment.

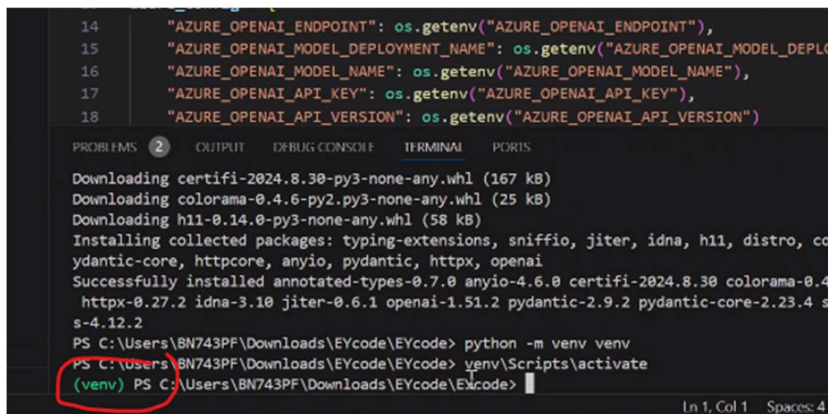
1. In VS Code, with the project root folder (genaitraining or code) selected, click on the +File icon to create a new file, name it “requirements.txt”
2. Put following content in requirements.txt and save the file (ctrl+s):

```
langchain-openai  
langchain-community  
langchain-text-splitters  
chromadb  
beautifulsoup4  
pypdf  
wikipedia  
plotly  
tqdm  
datasets  
python-dotenv  
openai  
streamlit  
dotted-dict
```

PyPDF2
matplotlib
seaborn
torch
transformers
tensorflow
tf-keras
jupyter
ipykernel
crewai
langgraph

3. Install packages from requirements.txt:

Make sure your virtual environment is active (you should see the venv name in the command prompt)



```
14 "AZURE_OPENAI_ENDPOINT": os.getenv("AZURE_OPENAI_ENDPOINT"),
15 "AZURE_OPENAI_MODEL_DEPLOYMENT_NAME": os.getenv("AZURE_OPENAI_MODEL_DEPLO
16 "AZURE_OPENAI_MODEL_NAME": os.getenv("AZURE_OPENAI_MODEL_NAME"),
17 "AZURE_OPENAI_API_KEY": os.getenv("AZURE_OPENAI_API_KEY"),
18 "AZURE_OPENAI_API_VERSION": os.getenv("AZURE_OPENAI_API_VERSION")

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Downloading certifi-2024.8.30-py3-none-any.whl (167 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
Installing collected packages: typing-extensions, sniffio, jiter, idna, h11, distro, co
ydantic-core, httpcore, anyio, pydantic, httpx, openai
Successfully installed annotated-types-0.7.0 anyio-4.6.0 certifi-2024.8.30 colorama-0.4
httpx-0.27.2 idna-3.10 jiter-0.6.1 openai-1.51.2 pydantic-2.9.2 pydantic-core-2.23.4 s
s-4.12.2
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate
(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode> |
```

Run the following command:

`pip install -r requirements.txt`

You may see error during the installation mentioning something like below:



```
running bdist_wheel
running build
running build_ext
building 'hnsplib' extension
error: Microsoft Visual C++ 14.0 or greater is required. Get it with "Microsoft C++ Build Tools": https://visualstudio.microsoft.com/visual-cpp-build-tools/
[end of output]

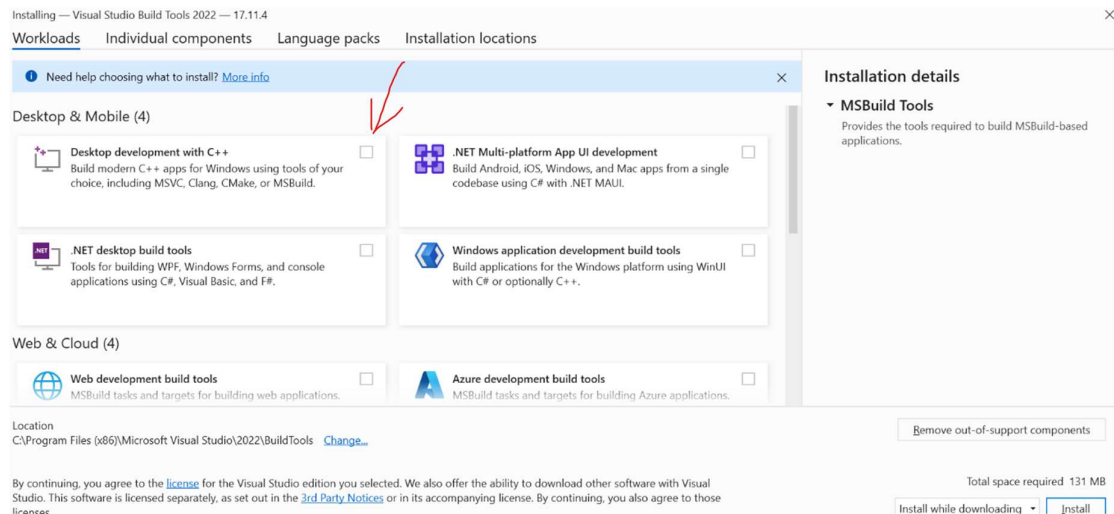
note: This error originates from a subprocess and is likely not a problem with pip.
```

To fix it, install MS dev tools:

<https://visualstudio.microsoft.com/visual-cpp-build-tools/>

Name	Date modified	Type
▼ Today (5)		
vs_BuildTools.exe	10-10-2024 02:32 PM	Application
tesla_db.zip	10-10-2024 12:24 PM	Compressed (zipped)
tesla-annual-reports.zip	10-10-2024 12:09 PM	Compressed (zipped)
index_tesla_documents.ipynb	10-10-2024 12:00 PM	Jupyter Source File
retrieval_tesla_documents.ipynb	10-10-2024 11:53 AM	Jupyter Source File
▼ Yesterday (6)		

When you open the installer, select the first box below:



After installation, restart VS Code

And open the code folder.

Reactivate the virtual environment in the terminal:

In terminal, execute the following:

```
➤ venv\Scripts\activate
```

Install packages again:

```
pip install -r requirements.txt
```

Troubleshooting

- If terminal says scripts are disabled (PowerShell):

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

- If Python is not recognized, verify it's added to PATH
- Make sure to always activate virtual environment before installing packages. The active environment is shown in green as below:

Give command:

`venv\Scripts\activate`

The screenshot shows a VS Code terminal window with a Python script being executed. The script prints a table of data for four cities: Delhi, Bangalore, Hyderabad, and Ahmedabad. Below the script output, the terminal shows the command `PS D:\code\EYcode> venv\Scripts\activate` being entered, and the prompt changes to `(venv) PS D:\code\EYcode>`, indicating the virtual environment is active. The prompt change is highlighted with a red circle.

```

45 # Create a twin axes for population and literacy rate
46 fig, ax1 = plt.subplots(figsize=(15, 8))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

1 Delhi 29000000 86.2
2 Bangalore 13000000 88.7
3 Hyderabad 9700000 83.3
4 Ahmedabad 8000000 89.1

PS D:\code\EYcode>
* History restored

PS D:\code\EYcode>
* History restored

PS D:\code\EYcode> venv\Scripts\activate
(venv) PS D:\code\EYcode>

```

- If still facing errors, try following and then run `pip install -r requirements.txt`:
`pip install --upgrade`
`pip setuptools wheel`
`pip cache purge`
`pip install --upgrade --force-reinstall setuptools`

Ater successful installation of packages you should see:

The screenshot shows a VS Code terminal window with the output of a `pip install` command. It lists the download of several wheels and the successful installation of various packages. Below the installation output, the terminal shows the command `PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv` being entered, followed by `PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate`, and the prompt changes to `(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>`, indicating the virtual environment is active.

```

13 azure_config = {
14     "AZURE_OPENAI_ENDPOINT": os.getenv("AZURE_OPENAI_ENDPOINT"),
15     "AZURE_OPENAI_MODEL_DEPLOYMENT_NAME": os.getenv("AZURE_OPENAI_MODEL_DEPLOYMENT_NAME"),
16     "AZURE_OPENAI_MODEL_NAME": os.getenv("AZURE_OPENAI_MODEL_NAME"),
17     "AZURE_OPENAI_API_KEY": os.getenv("AZURE_OPENAI_API_KEY"),
18     "AZURE_OPENAI_API_VERSION": os.getenv("AZURE_OPENAI_API_VERSION")
}

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

Downloading certifi-2024.8.30-py3-none-any.whl (167 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
Installing collected packages: typing-extensions, sniffio, jiter, idna, h11, distro, colorama, certifi, annot
ydantic-core, httpcore, anyio, pydantic, httpx, openai
Successfully installed annotated-types-0.7.0 anyio-4.6.0 certifi-2024.8.30 colorama-0.4.6 distro-1.9.0 h11-0.
httpx-0.27.2 idna-3.10 jiter-0.6.1 openai-1.51.2 pydantic-2.9.2 pydantic-core-2.23.4 sniffio-1.3.1 tqdm-4.66
s-4.12.2

PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate
(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>

```

4. Git

Git is a version control tool and is needed when interacting your app with Huggingface (HF) Huggingface requires your code to be checked into a HF backed git repository

1. Download Git

1. Visit the official Git website: <https://git-scm.com/download/windows>
2. The download should start automatically for the latest version
3. If it doesn't start automatically, click on the download link for Windows

2. Install Git

1. Run the downloaded installer (git-X.XX.X-64-bit.exe)
2. Adjust your PATH environment:
 - Select "Git from the command line and also from 3rd-party software" (recommended)
3. Complete the installation
4. Note the Path in which git is installed. You may need to add it to the System environment variables if you cannot run git from command prompt

3. Configure Git

1. Open Command Prompt or Git Bash
2. Set your username and email:

```
git config --global user.name "Your Name"
git config --global user.email "your email"
```

Note: When copying pasting the above commands, sometimes spurious characters are getting introduced. Hence manually type these commands if copy paste is not working.

4. Verify Installation

1. Open Command Prompt or Git Bash
2. Type:

```
git --version
```
3. You should see the Git version number if installation was successful

Troubleshooting

- If git commands aren't recognized, try restarting your command prompt or computer
- Ensure git is properly added to your PATH environment variable
- Check firewall settings if you have connectivity issues

For more detailed information and tutorials, visit the official Git documentation:
<https://git-scm.com/doc>

5. LM Studio

Participants should be able to download and install LM Studio on their PC:

<https://lmstudio.ai>

6. Claude Desktop (Optional)

Download and install Claude Desktop

<https://claude.ai/download>